(3)

(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION

(Programme: B. Tech. CSE-AI) (Semester: VI)

(March 2026) OFF LINE mode

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Subject Code: BAI 304	Subject: Deep Learning-II
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)
	a) Explain the effect of learning rate on the training of the Neural Network.	
	b) Discuss the concept of bias and variance with example.	
	c) Explain the sigmoid function along with advantages and disadvantages.	
	d) Discuss the concept of saddle point in detail with example.	
	UNIT I	
Q2	**Attempt any two parts**	*(5*2=10)*
	a) Explain the various strategies for weight initialization while training a neural network?	
	b) Discuss the concept of MP-Neuron along with its mathematical formulation. Identify the activation function used for this neuron.	
	c) Explain the concept of early stopping and batch normalization.	
	UNIT II	
Q3	**Attempt any two parts**	(5*2=10)
	a) Explain the RMPprop algorithm in detail. How is it different from Adaptive Gradient descent?	
	b) Differentiate the Stochastic Gradient Descent, Batch Gradient Descent, and Mini-Batch Gradient Descent methods. Discuss the mathematical formulation of each method.	
	c) Explain the concept of dropout layer with example. How is it different from L1/L2- regularization?	

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